**RAFFLES INSTITUTION**
**H2 Mathematics (9758)**
**2026 Year 6**

# Tutorial S2A: Discrete Random Variables

## Section A (Basic Questions)

**1** Find the mean and standard deviation of the random variable $X$ with the following probability distribution:

| $x$ | $-4$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|---|
| $\mathrm{P}(X = x)$ | 0.04 | 0.16 | 0.24 | 0.16 | 0.15 | 0.1 | 0.1 | 0.05 |

[ –0.83, 1.86]

**Solution:**

$X$ : denotes the random variable
$x$ : denotes the value that the r.v. can take

$$\mathrm{E}(X) = \sum x\,\mathrm{P}(X = x)$$

$$= (-4)(0.04) + (-3)(0.16) + (-2)(0.24) + (-1)(0.16) + 0 + 1(0.1) + 2(0.1) + 3(0.05) = -0.83$$

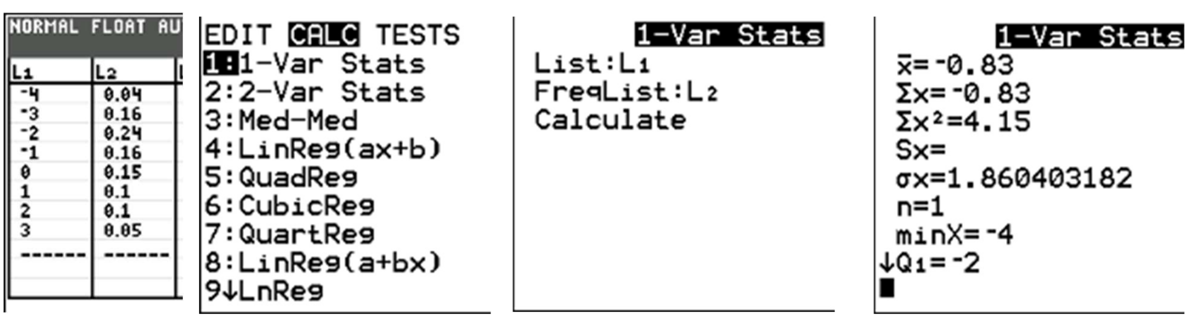
$$\mathrm{E}(X^2) = \sum x^2\,\mathrm{P}(X = x) = 4.15$$

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \left[\mathrm{E}(X)\right]^2$$

$$= 4.15 - (-0.83)^2 = 3.4611$$

Standard deviation $= \sqrt{\mathrm{Var}\,X} = 1.86$ (3 s.f)

**OR** Use GC. Enter data into 2 lists:




Read off the values from "sample mean", $\bar{x}$ and "sample standard deviation", $\sigma\ x$ .

E($X$) = –0.83
Standard deviation = 1.86 (3 s.f)

**2** Two dice are thrown and the numbers $A$ and $B$ shown on each die are noted.
The score $X$ from the throw is defined by:

$$X = \begin{cases} A+B & \text{if } A = B, \\ |A-B| & \text{if } A \neq B. \end{cases}$$

**(i)** Tabulate the probability distribution of $X$.

**(ii)** Evaluate the exact value of E($X$) and Var($X$).

[**(i)** $\frac{28}{9}$, **(ii)** $\frac{1015}{162}$]

**9233/1980/02/Q11 (modified)**

**Solution:**

**(i)**

| $x$ ($d_2$ across, $d_1$ down) | **1** | **2** | **3** | **4** | **5** | **6** |
|---|---|---|---|---|---|---|
| **1** | 2 | 1 | 2 | 3 | 4 | 5 |
| **2** | 1 | 4 | 1 | 2 | 3 | 4 |
| **3** | 2 | 1 | 6 | 1 | 2 | 3 |
| **4** | 3 | 2 | 1 | 8 | 1 | 2 |
| **5** | 4 | 3 | 2 | 1 | 10 | 1 |
| **6** | 5 | 4 | 3 | 2 | 1 | 12 |

Since each of the 36 possibilities are equally likely to occur, they each carry a probability of $\frac{1}{36}$. Probability distribution of $X$:

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 8 | 10 | 12 |
|---|---|---|---|---|---|---|---|---|---|
| $\mathrm{P}(X=x)$ | $\frac{10}{36}=\frac{5}{18}$ | $\frac{9}{36}=\frac{1}{4}$ | $\frac{6}{36}=\frac{1}{6}$ | $\frac{5}{36}$ | $\frac{2}{36}=\frac{1}{18}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ |

**(ii)**

$$\mathrm{E}(X) = \sum x\,\mathrm{P}(X=x)$$

$$= (1)\left(\frac{10}{36}\right)+(2)\left(\frac{9}{36}\right)+(3)\left(\frac{6}{36}\right)+(4)\left(\frac{5}{36}\right)+(5)\left(\frac{2}{36}\right)+(6)\left(\frac{1}{36}\right)+(8)\left(\frac{1}{36}\right)$$

$$+(10)\left(\frac{1}{36}\right)+(12)\left(\frac{1}{36}\right)=\frac{28}{9}$$

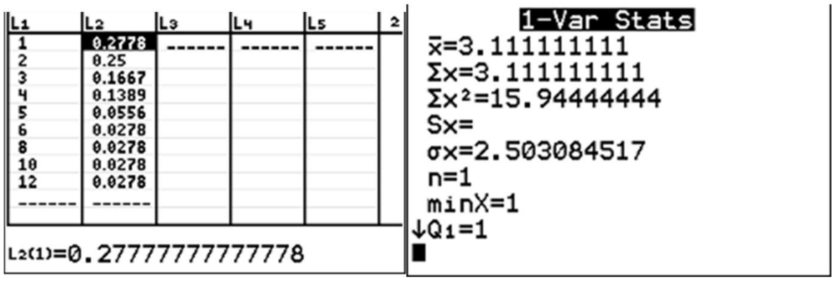
$$\mathrm{E}(X^2) = \sum x^2\,\mathrm{P}(X=x)$$

$$=(1)^2\left(\frac{10}{36}\right)+(2)^2\left(\frac{9}{36}\right)+(3)^2\left(\frac{6}{36}\right)+(4)^2\left(\frac{5}{36}\right)+(5)^2\left(\frac{2}{36}\right)+(6)^2\left(\frac{1}{36}\right)+(8)^2\left(\frac{1}{36}\right)$$

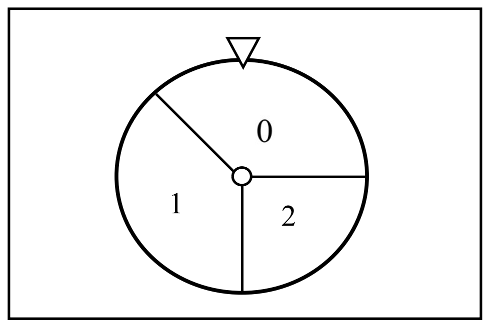
$$+(10)^2\left(\frac{1}{36}\right)+(12)^2\left(\frac{1}{36}\right)=\frac{287}{18}$$

$$\mathrm{Var}(X)=\mathrm{E}(X^2)-\left[\mathrm{E}(X)\right]^2$$
$$=\frac{287}{18}-\left(\frac{28}{9}\right)^2=\frac{1015}{162}$$

The following GC screenshots are only for the purpose of checking answers.
Note that the question asks for the exact values of E($X$) and Var($X$).




**3** A circular card is divided into 3 sectors scoring 0, 1, 2 and having angles $135°$, $135°$ and $90°$ respectively. The card is mounted onto a wall at its centre such that it can rotate freely. A pointer is fixed on the wall just above the card, as shown in the diagram.




In a game, a participant will spin the card twice, and the random variable $X$ is the product of the scores of the 2 spins.

**(i)** Tabulate the probability distribution of $X$. [3]

**(ii)** Show that $\mathrm{E}(X)=\frac{49}{64}$, and find the exact value of $\mathrm{Var}(X)$. [2]

Dave intends to use the above setup in a fundraising carnival. For every game, the participant will be charged \$$h$ to play, and will be awarded a prize money of \$$5X$ based on the outcome of their spins. Find the least integer value of $h$ such that on average, Dave can expect to earn at least \$1 from each participant. [2]

[**(ii)** $\frac{5343}{4096}$, 5]

**PJC Prelim 9758/2018/02/Q6 (modified)**

**Solution:**

**(i)**

$\mathrm{P}(\text{the score of one spin is } 0) = \mathrm{P}(\text{the score of one spin is } 1) = \frac{135}{360} = \frac{3}{8}$

$\mathrm{P}(\text{the score of one spin is } 2) = \frac{90}{360} = \frac{1}{4}$

$$\mathrm{P}(X=0) = \mathrm{P}(0,0) + \mathrm{P}(0,1)\times 2! + \mathrm{P}(0,2)\times 2!$$
$$= \left(\frac{3}{8}\right)^2 + \left(\frac{3}{8}\right)\left(\frac{3}{8}+\frac{1}{4}\right)2! = \frac{39}{64}$$
$$\mathrm{P}(X=1) = \mathrm{P}(1,1) = \left(\frac{3}{8}\right)^2 = \frac{9}{64}$$
$$\mathrm{P}(X=2) = \mathrm{P}(1,2)\times 2! = \frac{3}{8}\cdot\frac{1}{4}\cdot 2! = \frac{3}{16}$$
$$\mathrm{P}(X=4) = \mathrm{P}(2,2) = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$$

Note that since $\mathrm{P}(0,1) = \mathrm{P}(1,0)$, so
$\mathrm{P}(\text{one spin gives 0 and another gives 1})$
$= \mathrm{P}(0,1)\times 2!$

| $x$ | 0 | 1 | 2 | 4 |
|---|---|---|---|---|
| $\mathrm{P}(X=x)$ | $\frac{39}{64}$ | $\frac{9}{64}$ | $\frac{3}{16}$ | $\frac{1}{16}$ |

**(ii)**

$$\mathrm{E}(X) = (0)\left(\frac{39}{64}\right) + (1)\left(\frac{9}{64}\right) + (2)\left(\frac{3}{16}\right) + (4)\left(\frac{1}{16}\right) = \frac{49}{64} \text{ (shown)}$$
$$\mathrm{E}(X^2) = (0)^2\left(\frac{39}{64}\right) + (1)^2\left(\frac{9}{64}\right) + (2)^2\left(\frac{3}{16}\right) + (4)^2\left(\frac{1}{16}\right) = \frac{121}{64}$$
$$\mathrm{Var}(X) = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2 = \frac{5343}{4096}$$

Dave earns $\$(h-5X)$ from a participant.
$$\mathrm{E}(h-5X) \geq 1$$
$$h - 5\mathrm{E}(X) \geq 1$$
$$h \geq 1 + 5\mathrm{E}(X)$$
$$h \geq 1 + 5\times\frac{49}{64} = 4.8281$$
Hence, least integer value of $h$ is 5.

Recall: If $a$ and $b$ are constants, then $\mathrm{E}(a+bX) = a + b\mathrm{E}(X)$

***THE END***